

AI Engineer Foundation

Python for Data Science & Mathematics for AI

Mini-Project	Analyzing employee attrition factors, building mathematical insights, and processing HR data to support the AI pipeline
Time	2 weeks
Start	5 May 2026
End	15 May 2026

Dataset: IBM HR Analytics Employee Attrition & Performance

URL: [IBM HR Analytics Employee Attrition & Performance](#)

Dataset Description: This dataset is a fictional data set created by IBM data scientists to uncover the factors that lead to employee attrition. It contains **1,470 rows and 35 features** covering demographic information, job-related metrics, and satisfaction levels.

- **Target Variable:** *Attrition* (Yes/No) - indicates whether an employee has left the organization.
- **Subscribed Services/Metrics:** EnvironmentSatisfaction, JobInvolvement, JobLevel, JobSatisfaction, PerformanceRating, RelationshipSatisfaction, WorkLifeBalance.
- **Employee Information:** Age, BusinessTravel, Department, Education, EducationField, Gender, JobRole, MaritalStatus, OverTime.
- **Financial & Tenure Metrics:** MonthlyIncome, MonthlyRate, DailyRate, HourlyRate, PercentSalaryHike, StockOptionLevel, TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager.

MODULE 1: NUMPY (10 Questions)

- Q1.** Load the dataset into a NumPy array using **numerical features only** (Age, MonthlyIncome, TotalWorkingYears). Check the shape and dtype of the array.
- Q2.** Compute the Mean, Median, and Standard Deviation for the columns *Age* and *MonthlyIncome* using NumPy functions.
- Q3.** Apply Min-Max normalization to scale *MonthlyIncome* and *TotalWorkingYears* to the range [0,1] using vectorized operations.
- Q4.** Use boolean indexing to filter employees with *Age* > 35 and *MonthlyIncome* less than the average income of the dataset.
- Q5.** Compute the correlation matrix between *Age*, *MonthlyIncome*, and *YearsAtCompany* using NumPy (without using Pandas).

Vectorization & Linear Algebra

- Q6.** Implement a fully vectorized function to compute pairwise Euclidean distance between employees using the numerical features *MonthlyIncome* and *TotalWorkingYears* (avoid explicit Python loops).
- Q7.** Implement both Min-Max scaling and Z-score normalization from scratch using NumPy and compare the results for the *MonthlyIncome* column.
- Q8.** Compute cosine similarity between two employee vectors selected from the dataset (using Age, MonthlyIncome, and TotalWorkingYears).
- Q9.** Perform PCA manually using NumPy: construct the covariance matrix from the selected numerical features, compute eigenvalues and eigenvectors, and project the data onto the top 2 principal components.
- Q10.** Implement batch processing by splitting the NumPy array into equal-sized batches of 50 employees and computing the mean vector for each batch.

MODULE 2: MATHEMATICS FOR AI (10 Questions)

- Q1.** Represent the numerical features of the dataset as a matrix **X** and determine its shape and rank.
- Q2.** Encode the target variable *Attrition* (Yes=1, No=0), then apply Linear Regression using *Age* and *TotalWorkingYears* to predict attrition based on the model: $y = Xw + b$.
- Q3.** Compute the probabilities $P(\text{Attrition} = 1)$ and the conditional probability $P(\text{Attrition} = 1 \mid \text{OverTime} = \text{'Yes'})$ from the dataset.

- Q4.** Given the Mean Squared Error (MSE) loss, derive and compute the gradient with respect to $\mathbf{w}$ for the Linear Regression model: $L = (1/n) * \sum (y - \hat{y})^2$.
- Q5.** Compute the variance of *MonthlyIncome* and the covariance between *MonthlyIncome* and *YearsAtCompany*.
- Q6.** Construct the covariance matrix using numerical features (Age, Income, Tenure) and compute its eigenvalues and eigenvectors.
- Q7.** Perform Singular Value Decomposition (SVD) on matrix $\mathbf{X}$ and interpret the meaning of $\mathbf{U}$, $\mathbf{\Sigma}$, and $\mathbf{V}^T$ in the context of identifying hidden employee traits.
- Q8.** Implement Gradient Descent to optimize the Linear Regression parameters for predicting *MonthlyIncome* based on *Age* and visualize the loss convergence.
- Q9.** Discretize *MonthlyIncome* into bins (Low/Medium/High), then apply Bayes' Theorem to compute $P(\text{Attrition} \mid \text{Income_bin} = \text{'Low'})$.
- Q10.** Apply both L1 (Lasso) and L2 (Ridge) regularization to the regression model and compare their effects on the weights of features like *Age* and *MonthlyIncome*.

MODULE 3: PANDAS (10 Questions)

- Q1.** Load the IBM HR dataset into a Pandas DataFrame and display *head()*, *info()*, and *describe()* to understand the distribution of employee data.
- Q2.** Clean the dataset by checking for duplicate rows and handling any potential missing values in columns like *MonthlyIncome* or *YearsAtCompany*.
- Q3.** Group the data by *Department* and *JobRole*, then compute the average monthly income and the attrition rate for each group.
- Q4.** Create a new feature *IncomePerYear = MonthlyIncome / TotalWorkingYears* (handle cases where *TotalWorkingYears* is zero) and analyze its correlation with *Attrition*.
- Q5.** Visualize the distribution of *Age* using a histogram and create a bar chart showing *Attrition* counts across different *BusinessTravel* frequencies.
- Q6.** Create a pivot table with *JobRole* as the index, *Attrition* as columns, and the average *JobSatisfaction* as values.
- Q7.** Create *TotalWorkingYears* groups (e.g., 0-5, 5-10, 10-20, 20+) and analyze how attrition rate changes as employees gain more experience.
- Q8.** Segment employees into 'High Earners' and 'Low Earners' based on the median *MonthlyIncome* and compare their *WorkLifeBalance* scores.

Q9. Generate advanced visualizations: a correlation heatmap for all numerical features and a boxplot of *MonthlyIncome* grouped by *Attrition* status.

Q10. Perform feature engineering: apply One-Hot Encoding to *Department* and *EducationField*, and label encode *OverTime* to prepare the dataset for a machine learning model.

Submission Requirements and Implementation Guidelines

Students are required to complete the mini-project using Python, implemented either in LMS Quanskill, VS Code (local environment), or Google Colab (cloud environment). The submission must demonstrate a complete and reproducible workflow from data loading to analysis and results.

Each submission should include: a well-structured Python script or notebook (.py or .ipynb) with clear organization of code into logical sections (e.g., data loading, preprocessing, analysis, visualization); appropriate use of libraries such as NumPy, Pandas, and Matplotlib/Seaborn; and meaningful comments explaining key steps and decisions.

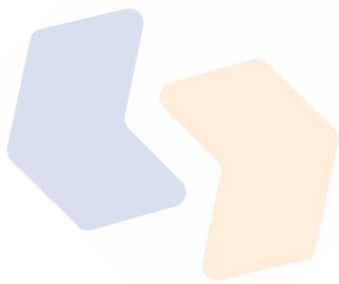
Students must ensure that the code runs without errors and produces all required outputs (tables, statistics, and visualizations). In addition, a brief report (or markdown cells in Colab) should summarize the main findings, insights, and conclusions derived from the dataset.

All files should be submitted in a single package, including the source code and any necessary supporting files or instructions to reproduce the results.

Example Directory Structure

```
HR-analytics-project/
├── data/
│   ├── raw/
│   │   └── hr.csv
│   └── processed/
├── notebooks/
│   └── hr_analysis.ipynb
├── src/
│   ├── numpy_tasks.py
│   ├── math_tasks.py
│   ├── pandas_tasks.py
│   └── utils.py
├── outputs/
│   ├── figures/
│   ├── tables/
│   └── results.txt
└── report/
```

```
| └─ final_report.pdf
| └─ requirements.txt
| └─ README.md
```



Quanskill